(New Delhi, India) it may not be practically helpful. If we ask the relation in the following set of ordered pairs,

{(New Delhi, India), (Washington, USA), (Beijing, China), (London, U.K.), (Kathmandu, Nepal)} then specifying the relation is easy.

Progress Check

Let A = {1, 2, 3, 4} and B = {a, b, c}.

1. Which of the following are relations from A to B?	2. Which of the following are relations from B to A?
(i) { (1, b), (1, c), (3, a), (4, b) }	(i) { (c, a), (c, b), (c, 1) }
(ii) { (1, a), (b, 4), (c, 3) }	(ii) { (c, 1), (c, 2), (c, 3), (c, 4) }
(iii) { (1, a), (a, 1), (2, b), (b, 2) }	(iii) { (a, 4), (b, 3), (c, 2) }

Illustration 4

Students in a class	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_{10}
Heights (in feet)	4.5	5.2	5	4.5	5	5.1	5.2	5	4.7	4.9

Let us define a relation between heights of corresponding students. (Fig.1.7)

R = {(heights, students)}

$R = \{(4.5, S_1),\ (4.5, S_4),\ (4.7, S_9),\ (4.9, S_{10}),\ (5, S_3),\ (5, S_5), (5, S_8),\ (5.1, S_6),\ (5.2, S_2),\ (5.2, S_7)\}$

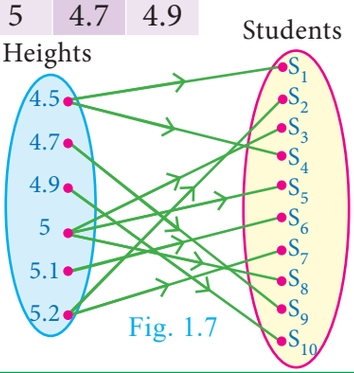


Fig. 1.7

Definition

Let A and B be any two non-empty sets. A 'relation' R from A to B is a subset of $A \times B$ satisfying some specified conditions. If $x \in A$ is related to $y \in B$ through R, then we write it as $x\,\mathrm{R}\,y$. $x\,\mathrm{R}\,y$ if and only if $(x, y) \in \mathrm{R}$.

The domain of the relation $\mathrm{R} = \{x \in A \mid x\,\mathrm{R}\,y, \text{for some } y \in B\}$

The co-domain of the relation R is B

The range of the relation $\mathrm{R} = \{y \in B \mid x\mathrm{R}y, \text{for some } x \in A\}$

From these definitions, we note that domain of $\mathrm{R} \subseteq A$, co-domain of $\mathrm{R} = B$ and range of $\mathrm{R} \subseteq B$.

Illustration 5

Let A = {1,2,3,4,5} and B = {Mathi, Arul, John}

A relation R between the above sets A and B can be represented by an arrow diagram (Fig. 1.8).

Then, domain of R = {1,2,3,4}

range of R= {Mathi, Arul, John} = co-domain of R.

Note that domain of R is a proper subset of A.

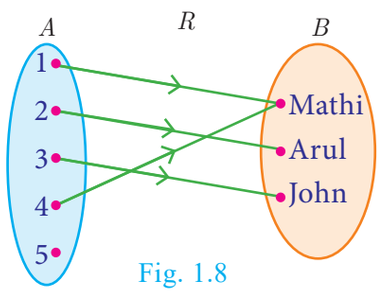


Fig. 1.8

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